

DÉTECTION NON SUPERVISÉE DE VOCALISATIONS DE CÉTACÉS PAR FACTORISATION DE MATRICES NON NÉGATIVES

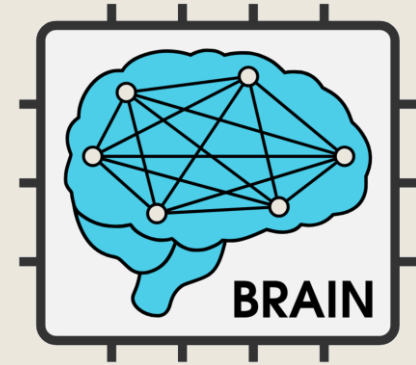
Mahamat Nour Bachar Mahamat, Axel Marmoret, Dorian Cazau

24 Juin 2026



Hello!

- Collaboration between
 - Mahamat Nour Bachar Mahamat (M2 intern)
 - Axel Marmoret (Assistant Professor, IMT Atlantique)
 - Dorian Cazau (Assistant Professor, ENSTA)
- Main motivation for the internship:
 - Applying a standard method in **Music** Event Detection in the context of underwater bioacoustics, and call detection



IMT Atlantique

Bretagne-Pays de la Loire
École Mines-Télécom

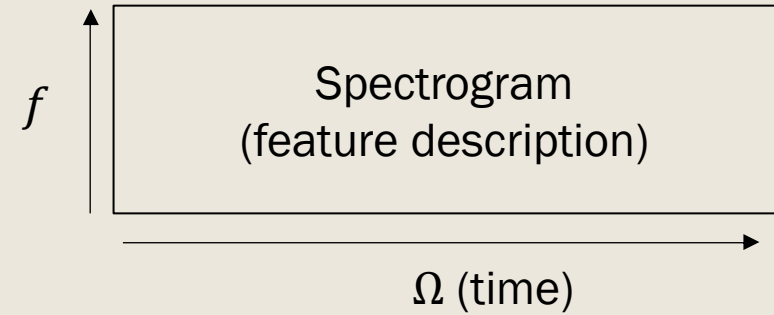
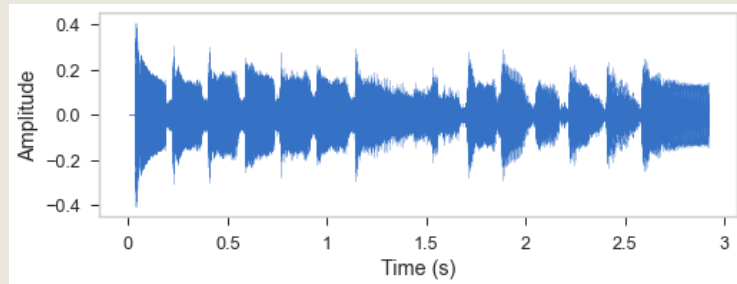


Detection of whale calls

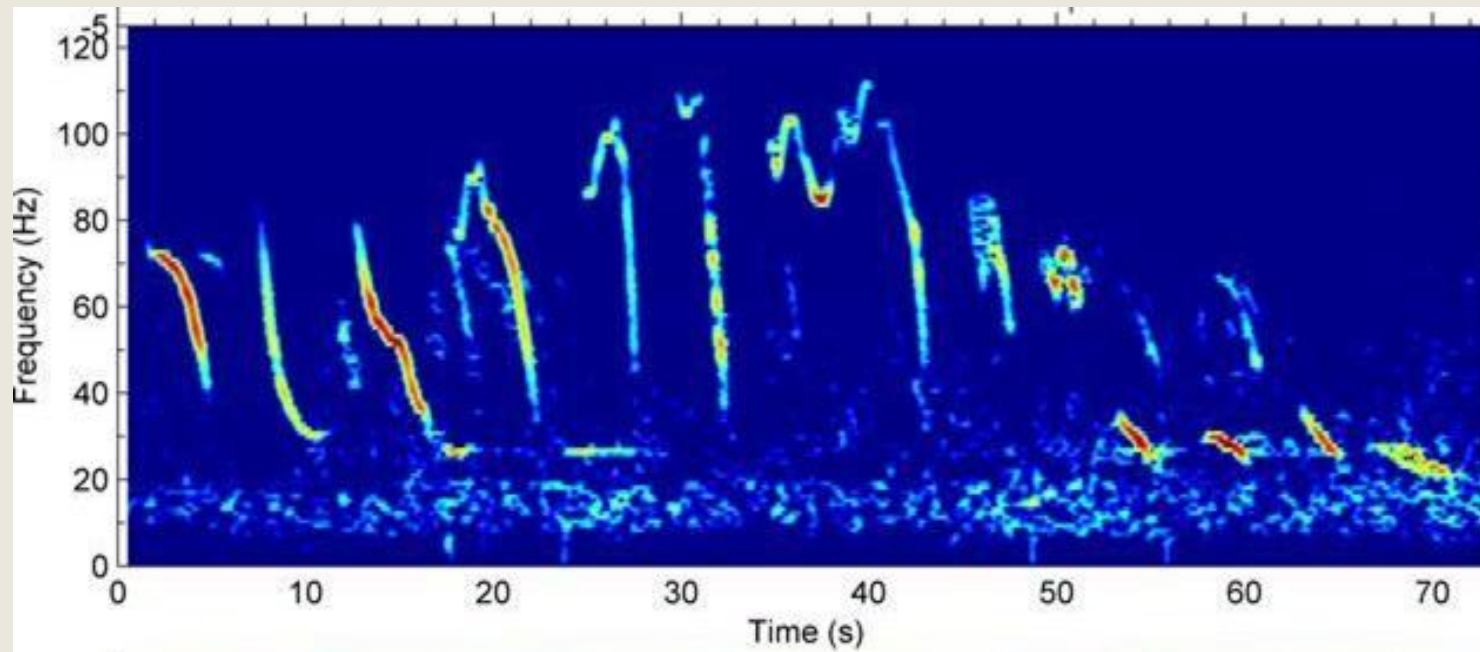
BioDCASE challenge:

<https://biodcase.github.io/challenge2025/task2>

Audio signal: spectrogram



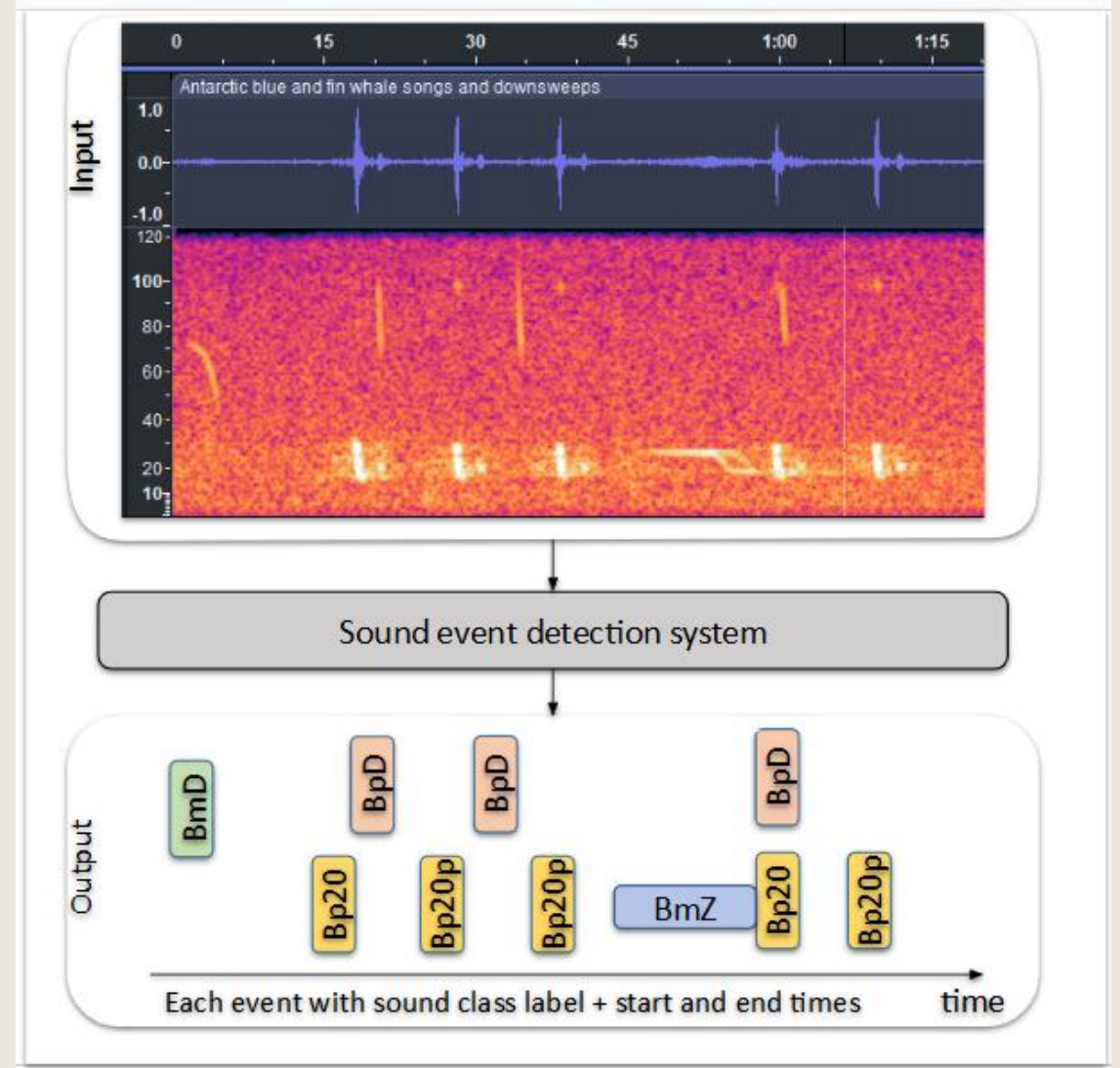
- Practically:



Detection of whale calls

- Detecting:
 - Which type of call is emitted
 - When the calls are emitted

➡ from the spectrogram



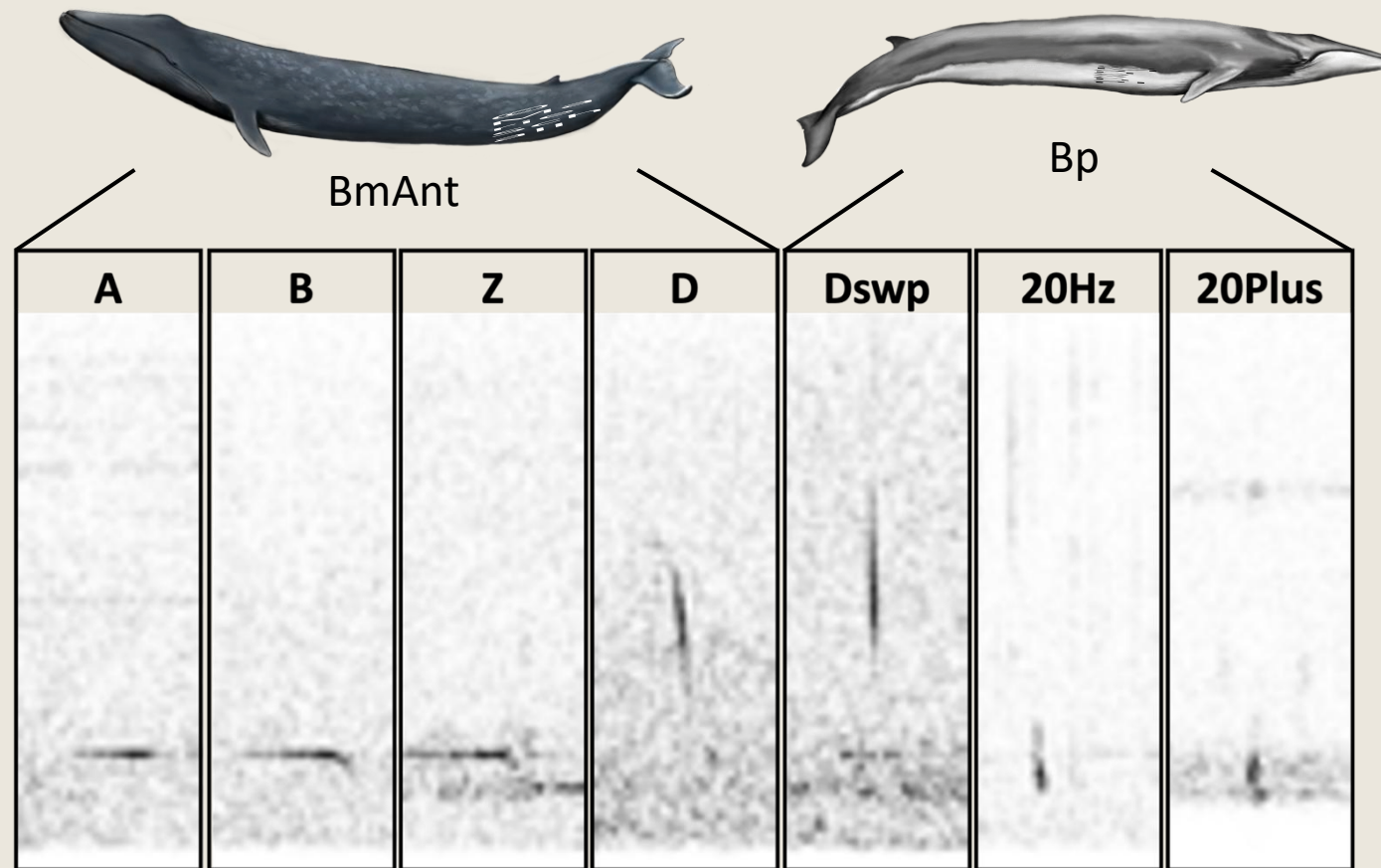
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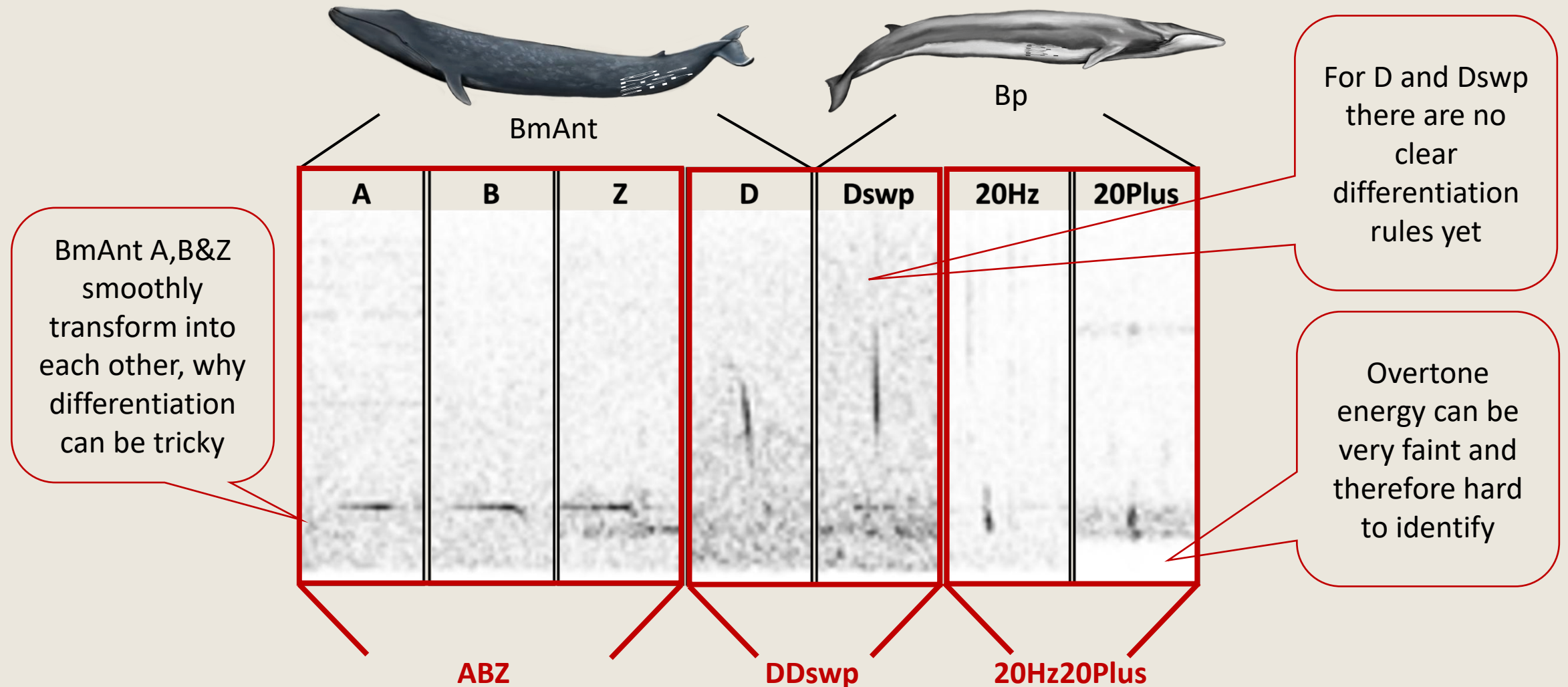
Type of calls:

7 annotation categories:

-> 4 Antarctic blue whale (BmAnt) and 3 fin whale (Bp)



But... Only 3 in practice here



Nonnegative Matrix Factorization (NMF)

[LS99]

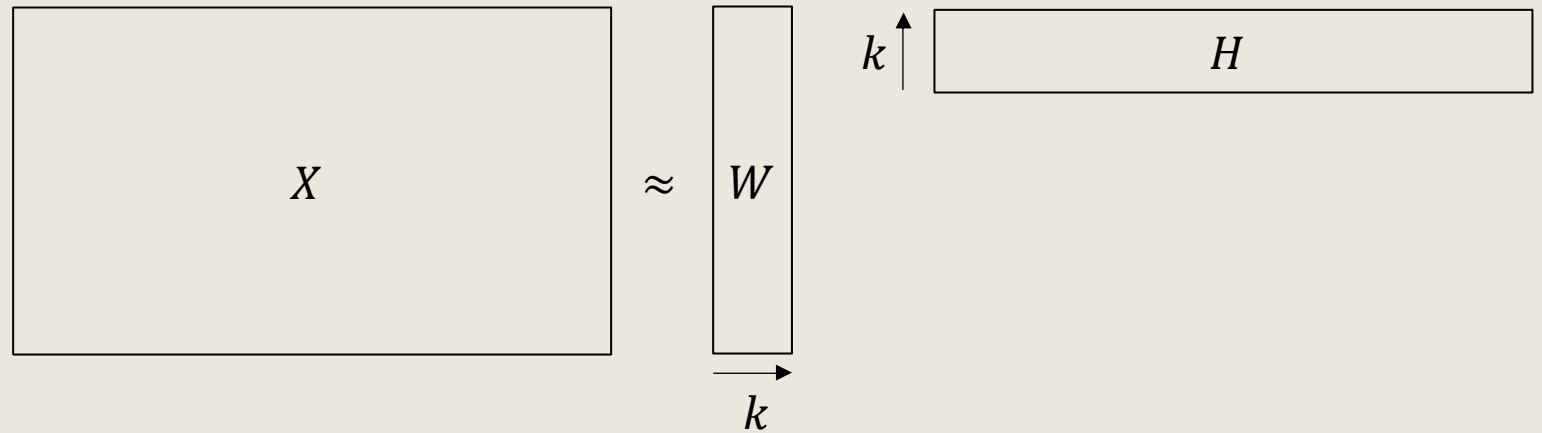
[LS99] Lee, D. D., & Seung, H. S. (1999). Learning the parts of objects by non-negative matrix factorization. *Nature*, 401(6755), 788-791.

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$$X \approx WH$$



$$\operatorname{argmin}_{W \geq 0, H \geq 0} \|X - WH\|_2$$

Why using NMF?

Practical advantages

- Unsupervised
- Interpretable
- Lightweight,
- Computationally cheap

Huge literature

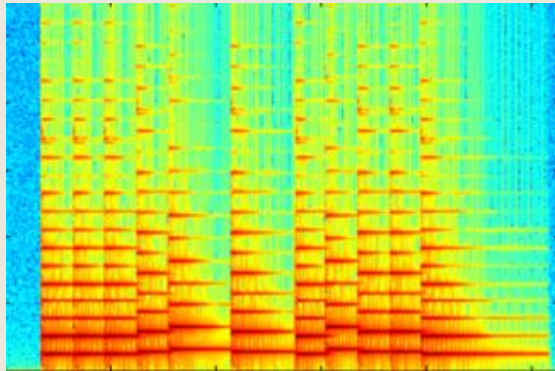
- 20 years of development in music processing (and, to some extent, speech)

Some works in underwater acoustics:

- Audio event detection [LT18, Lin21] and Sound classification [Zhe20]
- Blind Source Separation [Li23, LBP25, LSCB25]
- Denoising [JL21]

NMF for Music Transcription [SB03]

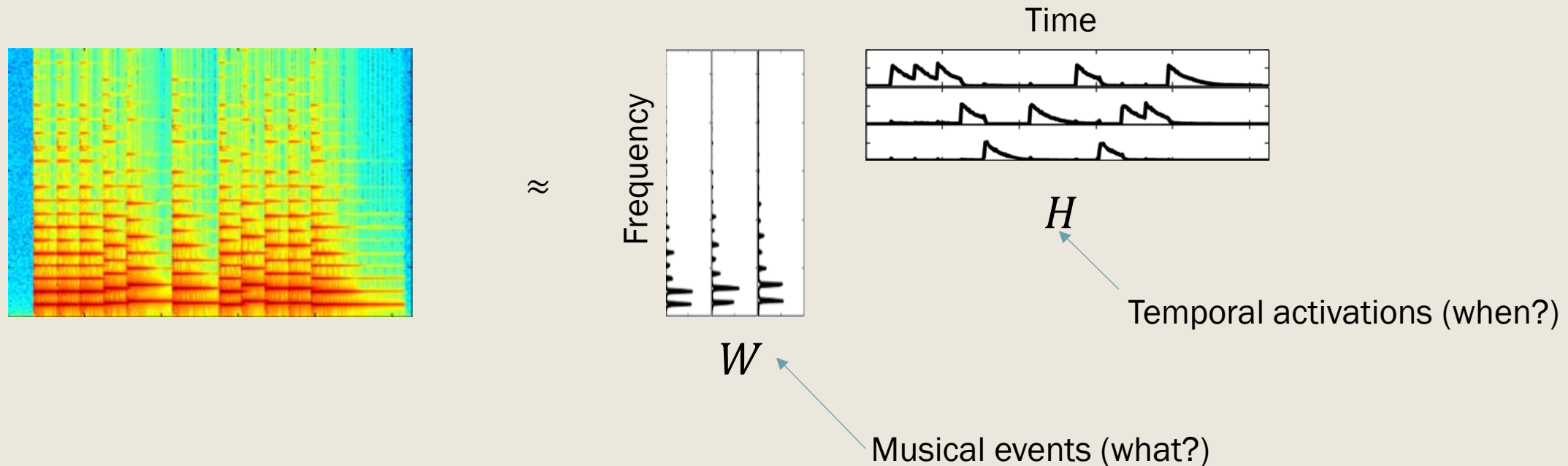
Example adapted from [Wu+22]



$$X \approx WH$$

NMF for Music Transcription [SB03]

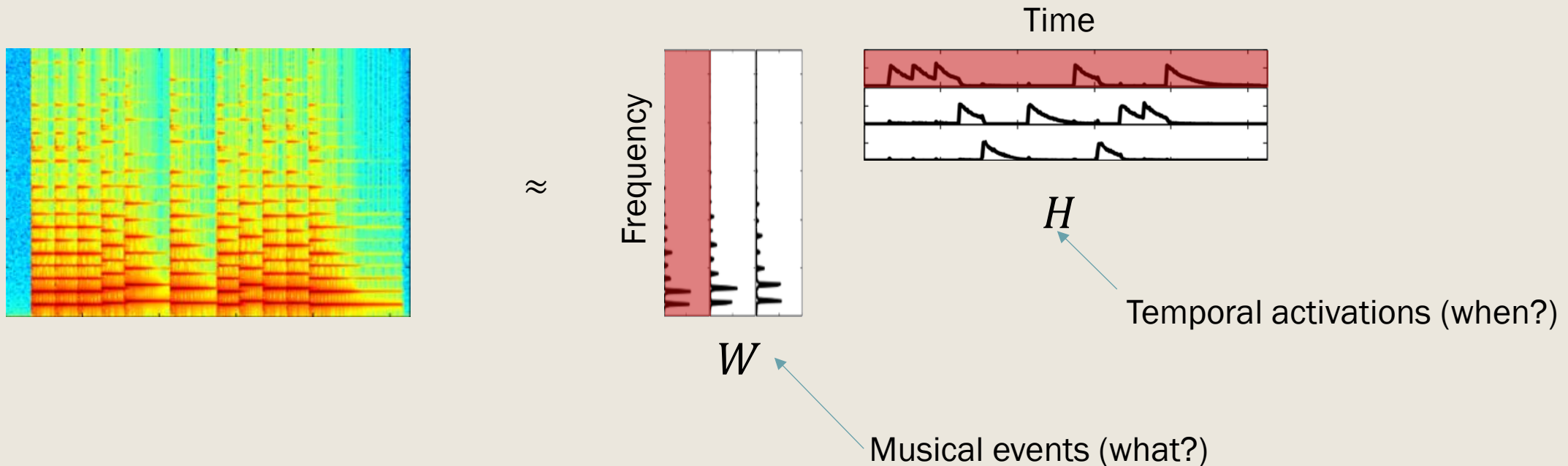
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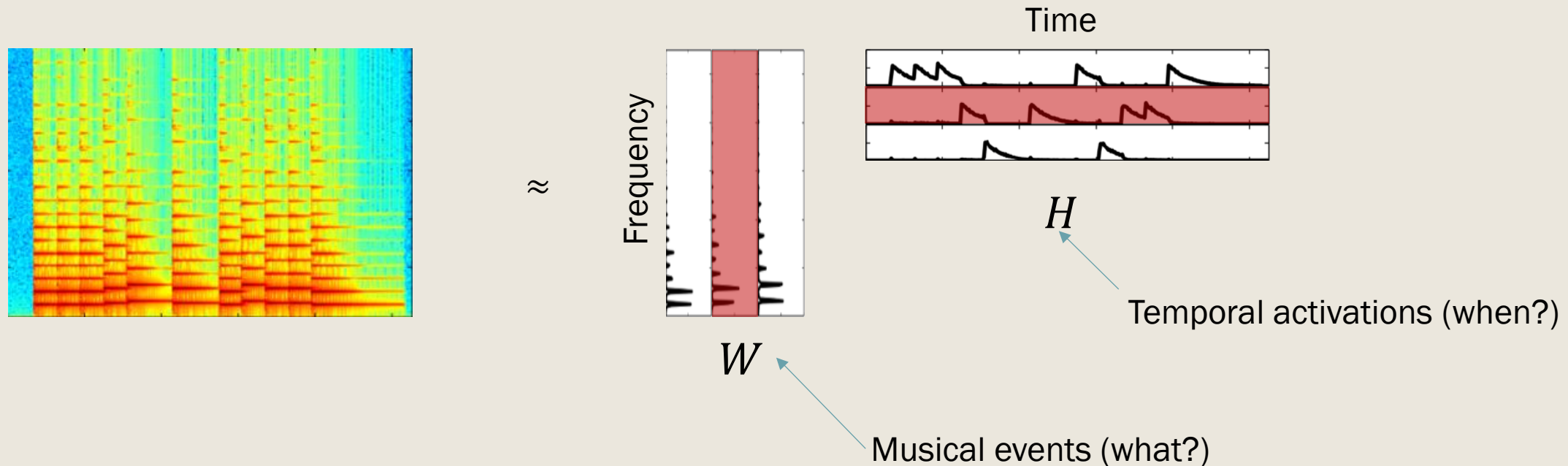
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NMF for Music Transcription [SB03]

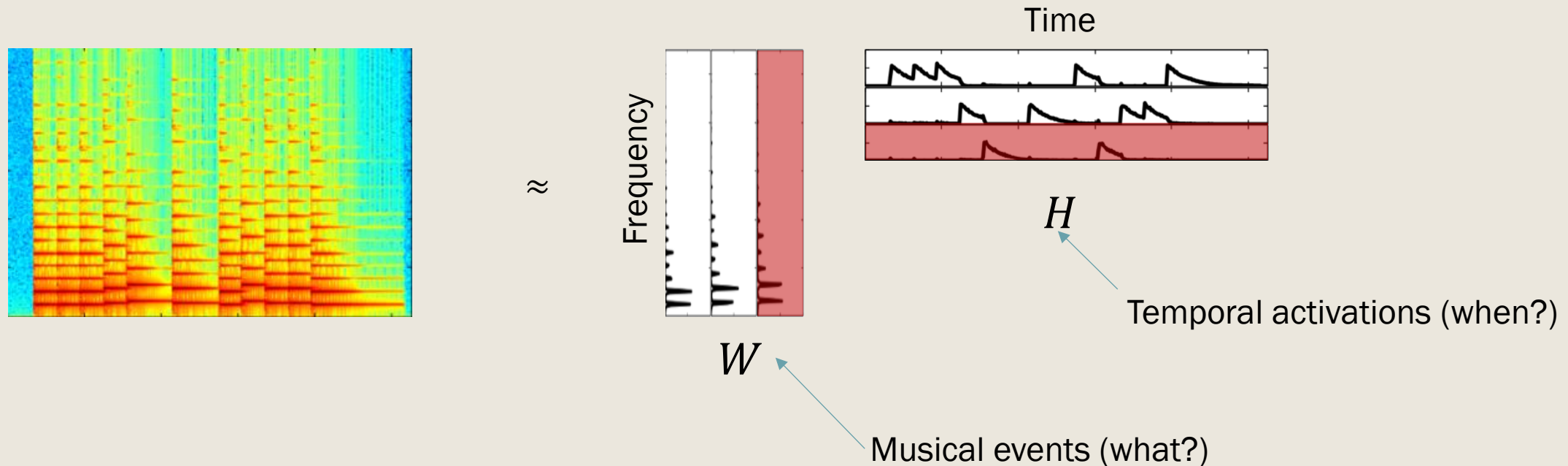
Example adapted from [Wu+22]



$$X \approx WH$$

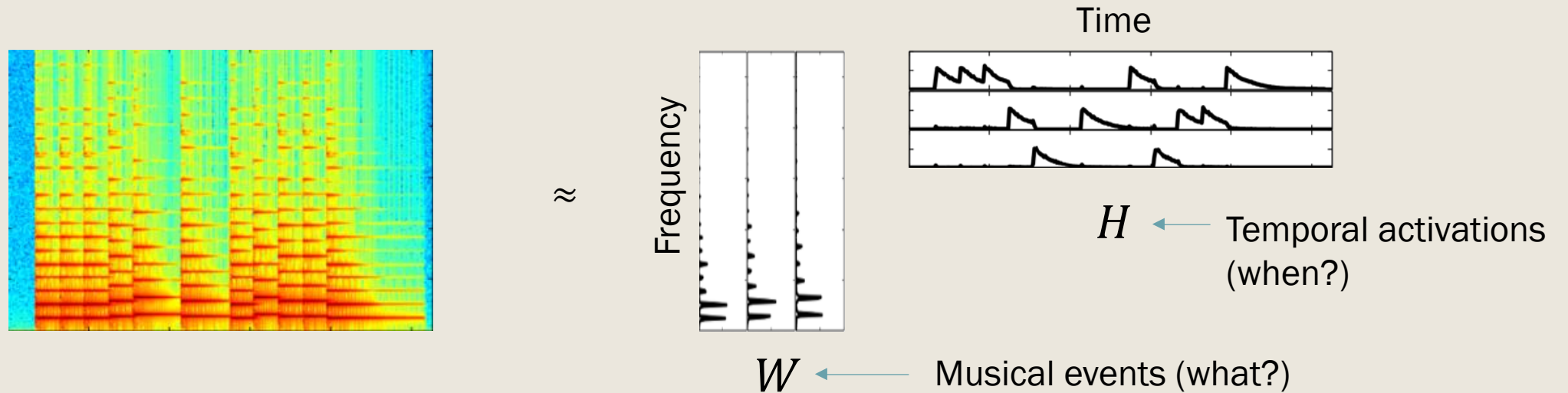
NMF for Music Transcription [SB03]

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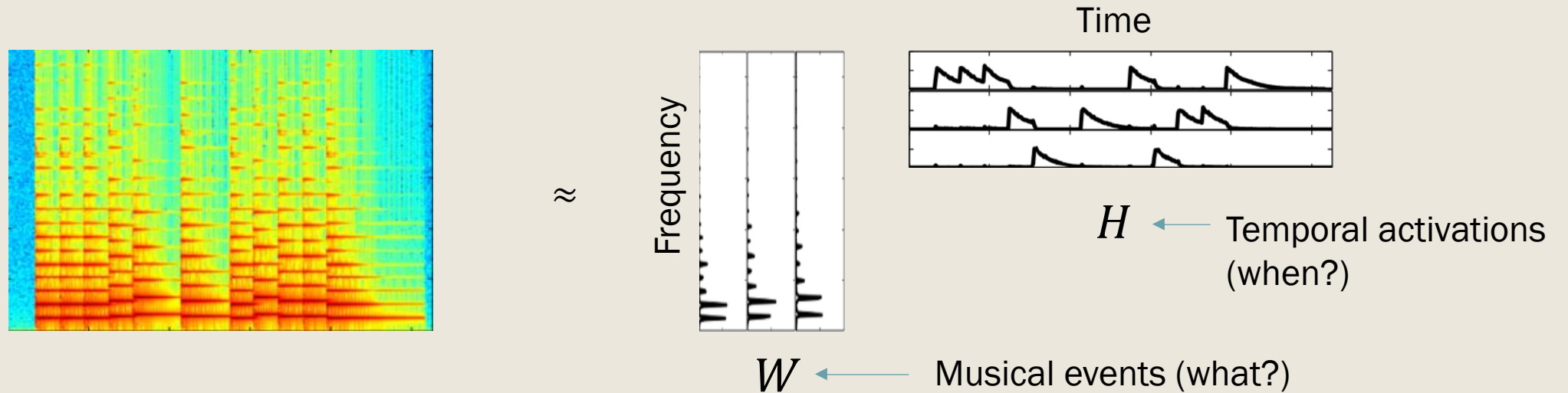
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NMF main assumptions



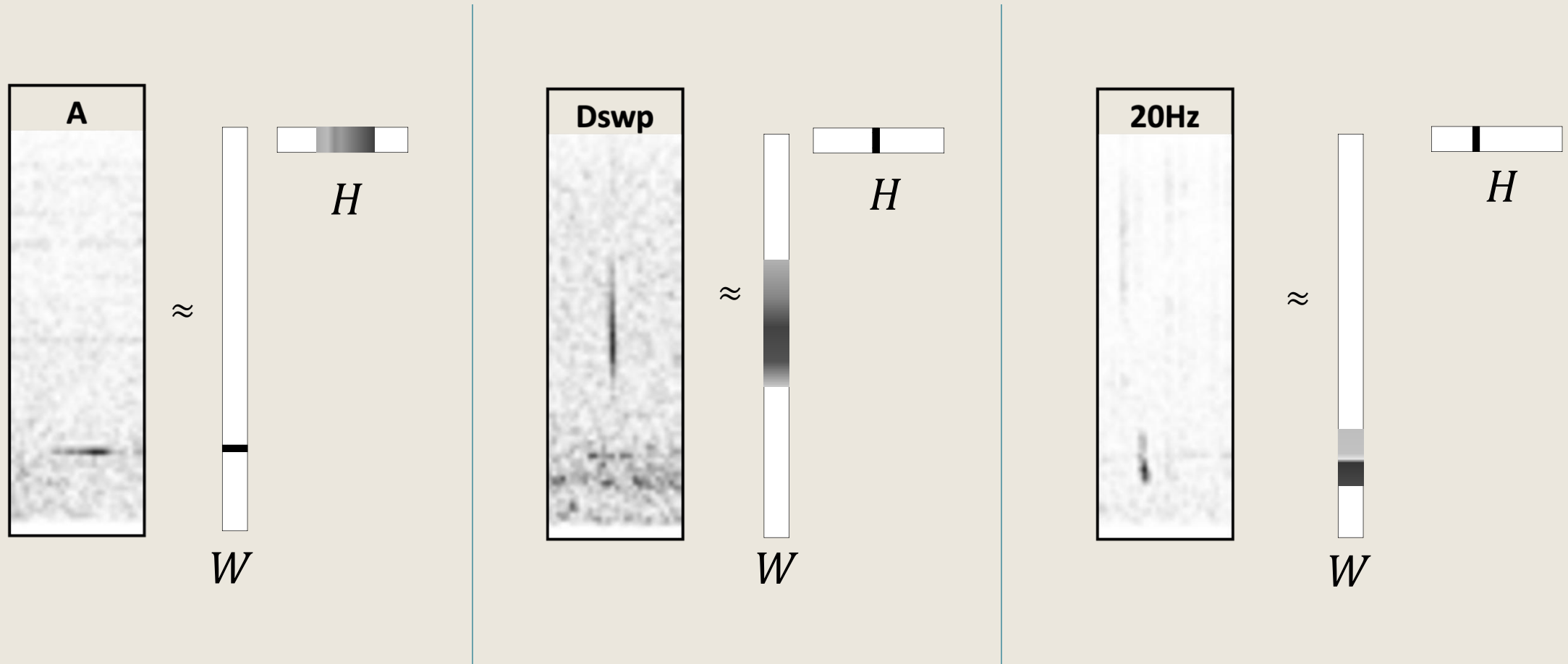
1. Events may be represented as a **one column vector** in the frequency domain

NMF main assumptions

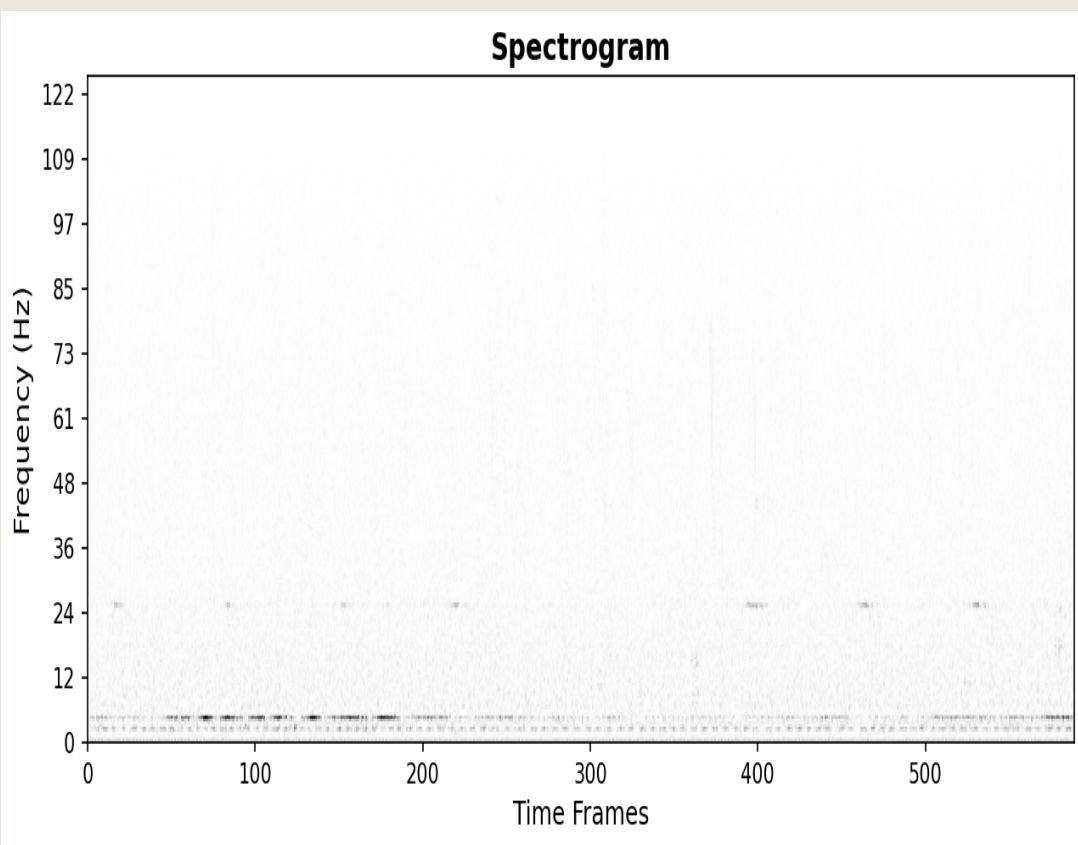


1. Events may be represented as a **one column vector** in the frequency domain
2. Events are **repeated several times** in the spectrogram

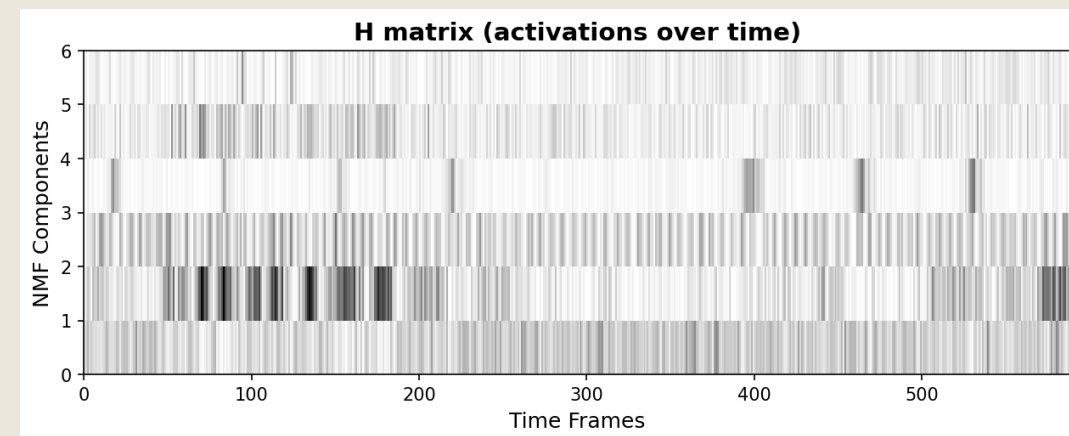
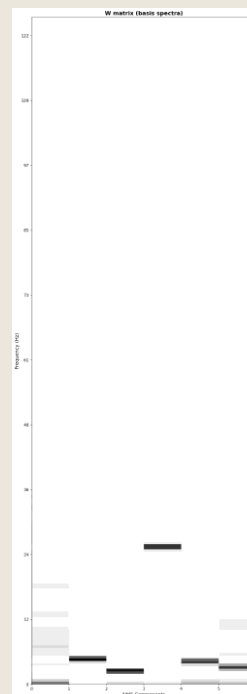
NMF and the three calls (ideally)



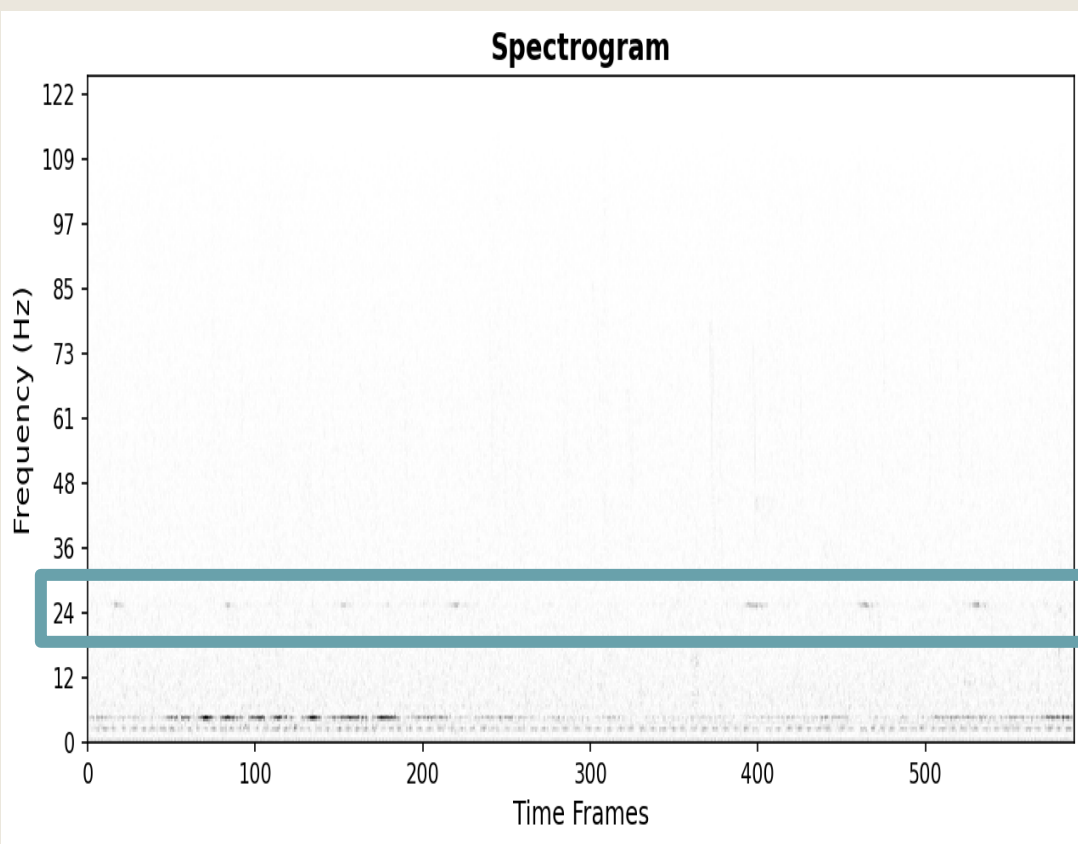
A practical example



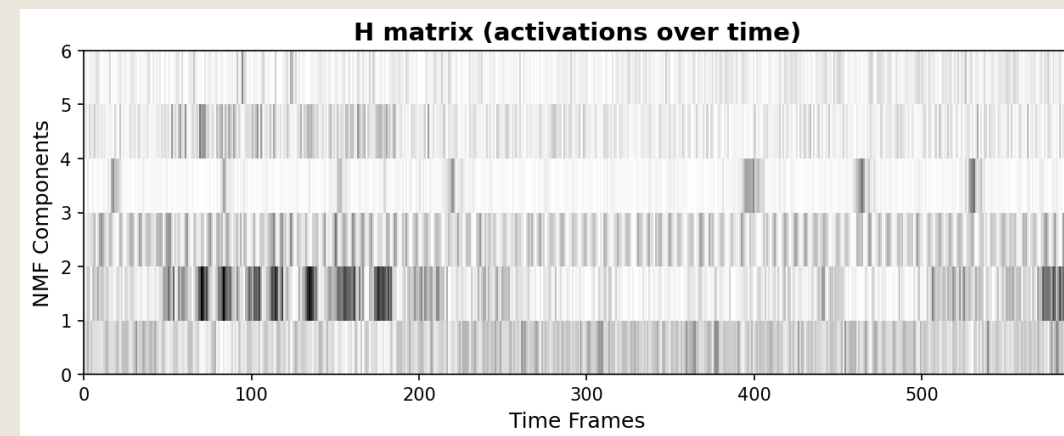
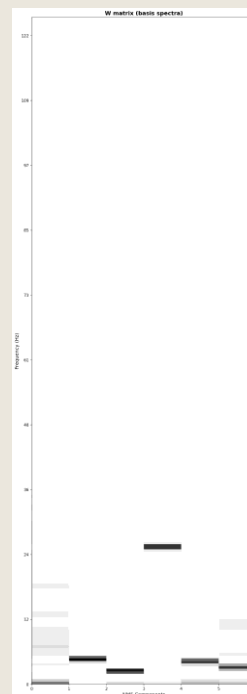
$\approx$



A practical example

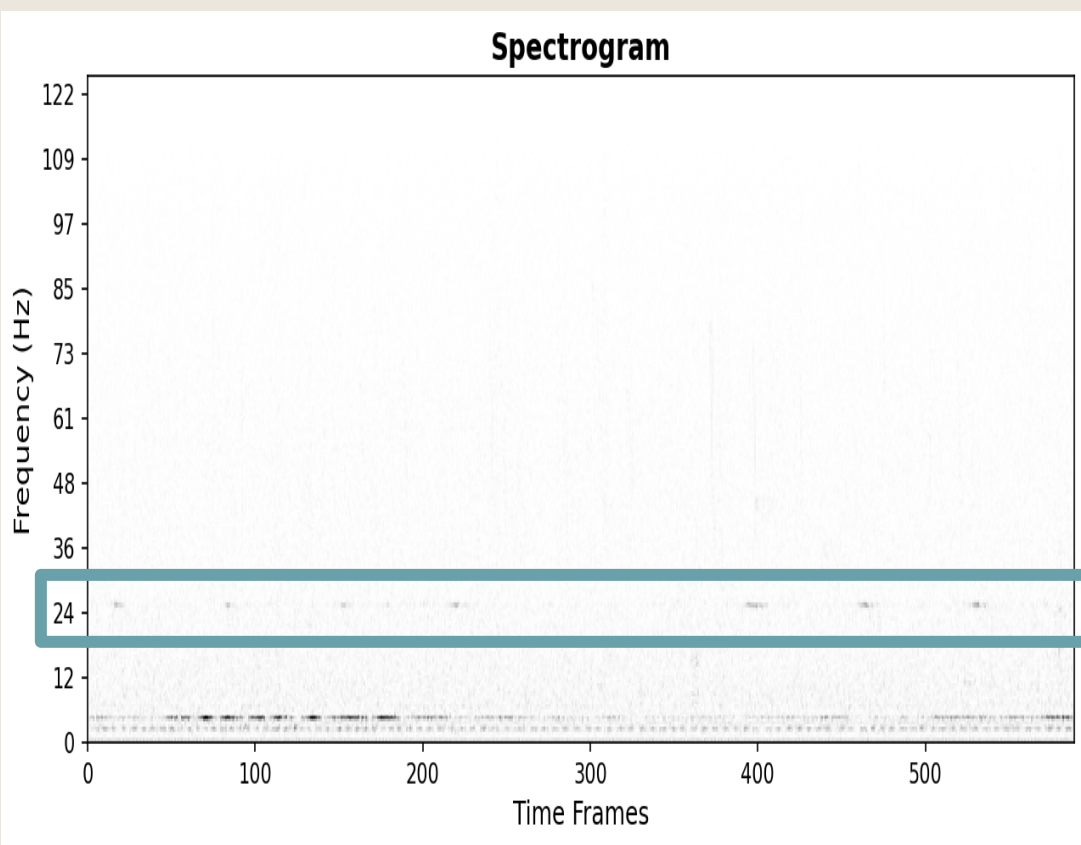


$\approx$

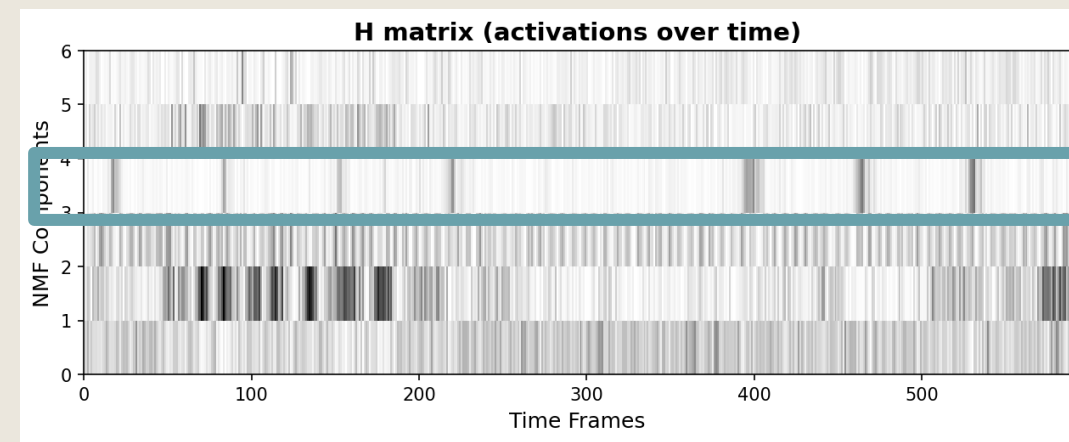
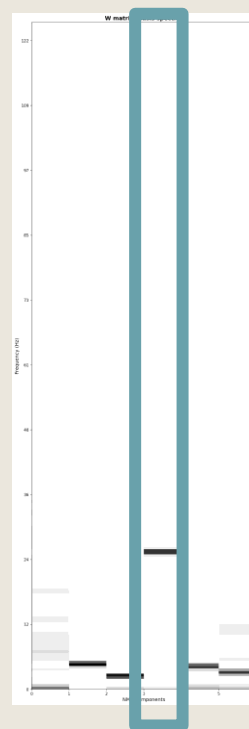


BmA call

A practical example

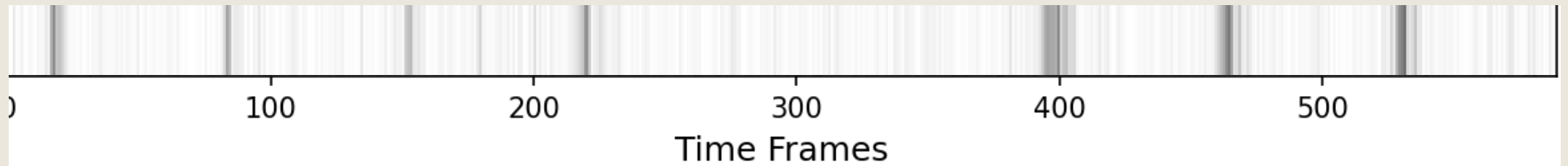


$\approx$



Post-processing H into detections

- Objective: transforming raw time series into calls onsets and offsets



`[(15.35, 22.60), (83.90, 91.27), (153.56, 157.74), (221.02, 227.16),
(401.14, 409.620), (469.01, 477.48), (537.97, 545.83)]`

Thresholding H

- Naïve and standard method: thresholding activations
 - *Simple but actually efficient*



- We could do better!
 - *WIP*

Experimental results

Experimental results

- Same protocol as [BioDCASE 2025 task 2](#)
 - *Detection considered valid if IoU $\geq 30\%$*
 - *Precision, Recall, and F1-score are derived from detections*
 - *Evaluated sets are Kerguelen2020 and ddu2021*
- Hyperparameters are selected based on the validation set

	ddu2021	Kerguelen2020
F1-scores	12,50%	39,47%

Comparison with BioDCASE results

Submission information		Rank		Dataset metrics		
Rank	Name	Technical Report	Official rank	Rank score	DDU2021 F-Score	Kerguelen2020 F-Score
1	Swin transformer fusion with softmax		1	50.00	46.40	50.40
2	Swin transformer fusion with softmax		2	43.80	38.80	53.40
3	Baseline YOLOv11		3	37.00	37.00	41.00
4	Whale-VAD		4	35.70	33.90	36.30
5	Whale-VAD		5	34.90	34.20	32.10
6	Whale-VAD		6	33.60	33.90	31.90
7	tiny yolov4 coco		7	27.50	27.50	28.90
NMF					12,50%	39,47%
8	voxaboxen dv		8	19.20	20.50	11.20
9	voxaboxen dv		9	17.80	18.50	14.10
10	voxaboxen dv		10	15.90	16.40	13.20

<https://biодcase.github.io/challenge2025/task2-results>

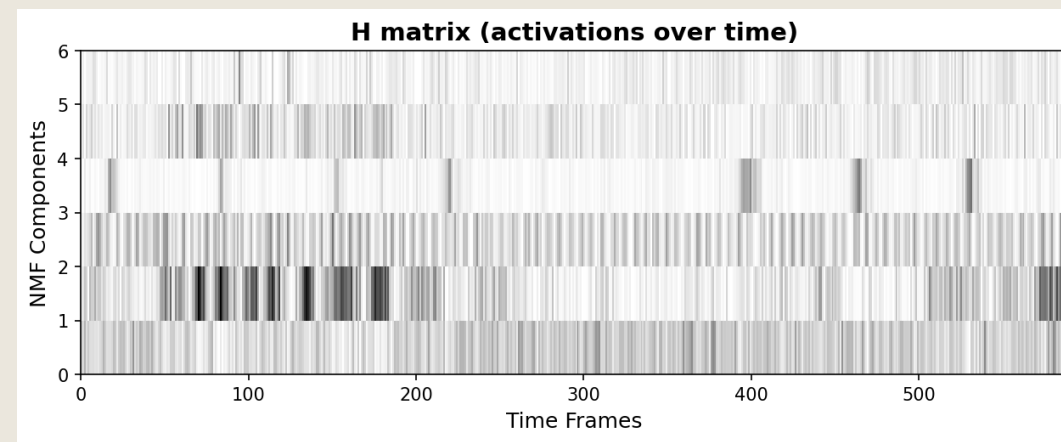
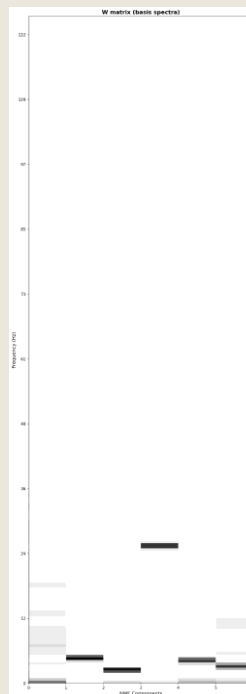
Experimental results

- Same protocol as [BioDCASE 2025 task 2](#)
 - Detection considered valid if IoU $\geq 30\%$
 - Precision, Recall, and F1-score are derived from confusion matrix
 - Evaluated sets are Keras and TensorFlow



NMF is unsupervised.

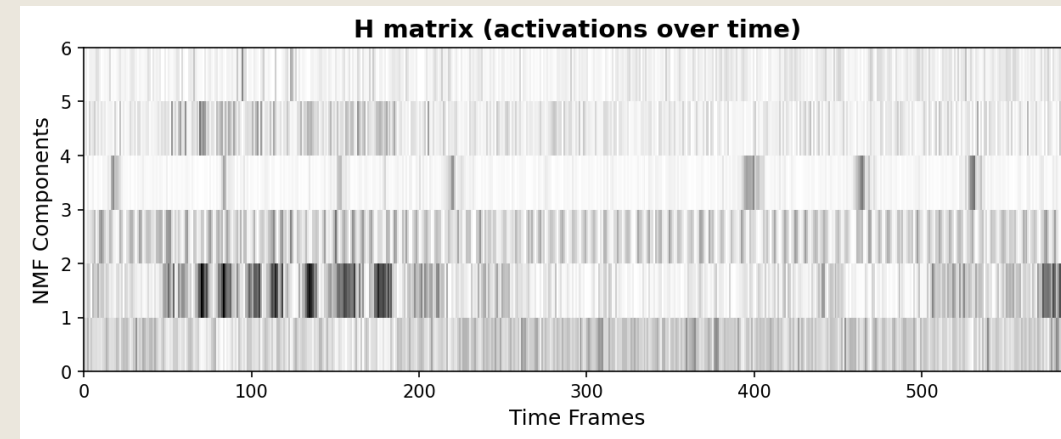
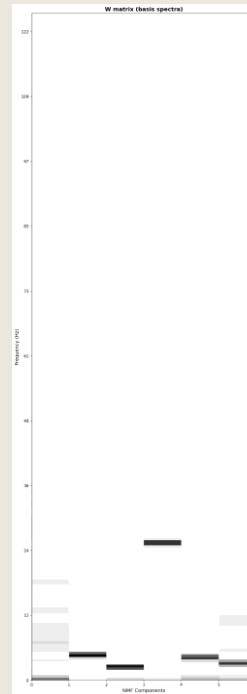
How can we associate a column to an event?



NMF is unsupervised.

How can we associate a column to an event?

Option 1: Manual inspection



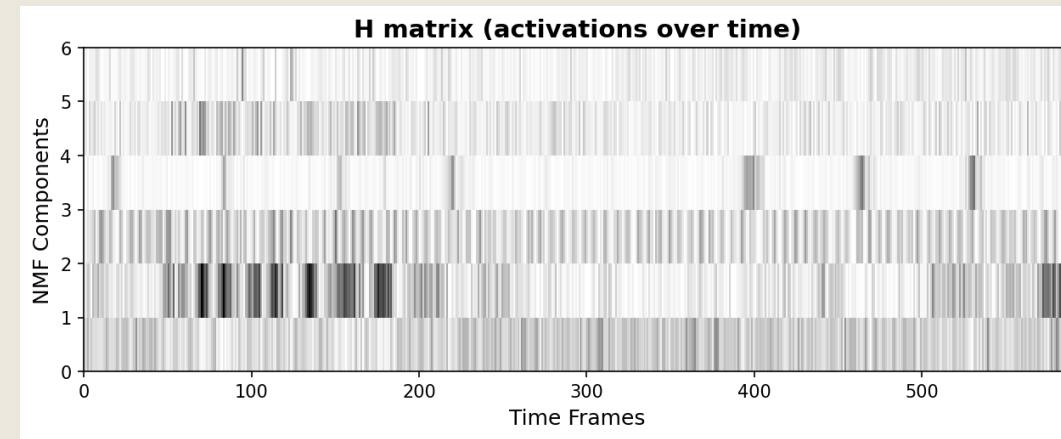
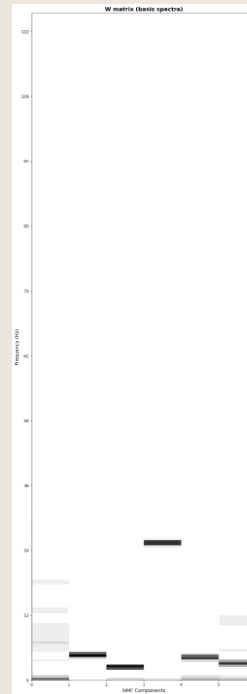
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Option 2: Best matching
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(requires annotations!)

➡ What was used for the previous results



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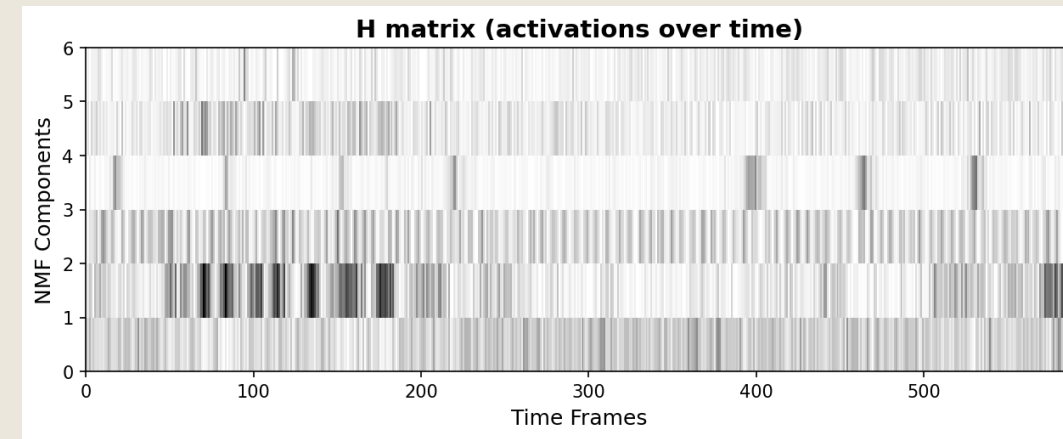
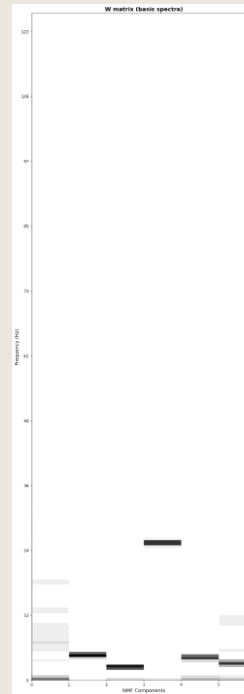
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➡ What was used for the previous results

Option 3: constraining W to
correspond to calls with prior
knowledge



The 3 calls of interest, again

We know the frequency bands of the calls we're detecting:



➡ Between 16 & 27 Hz



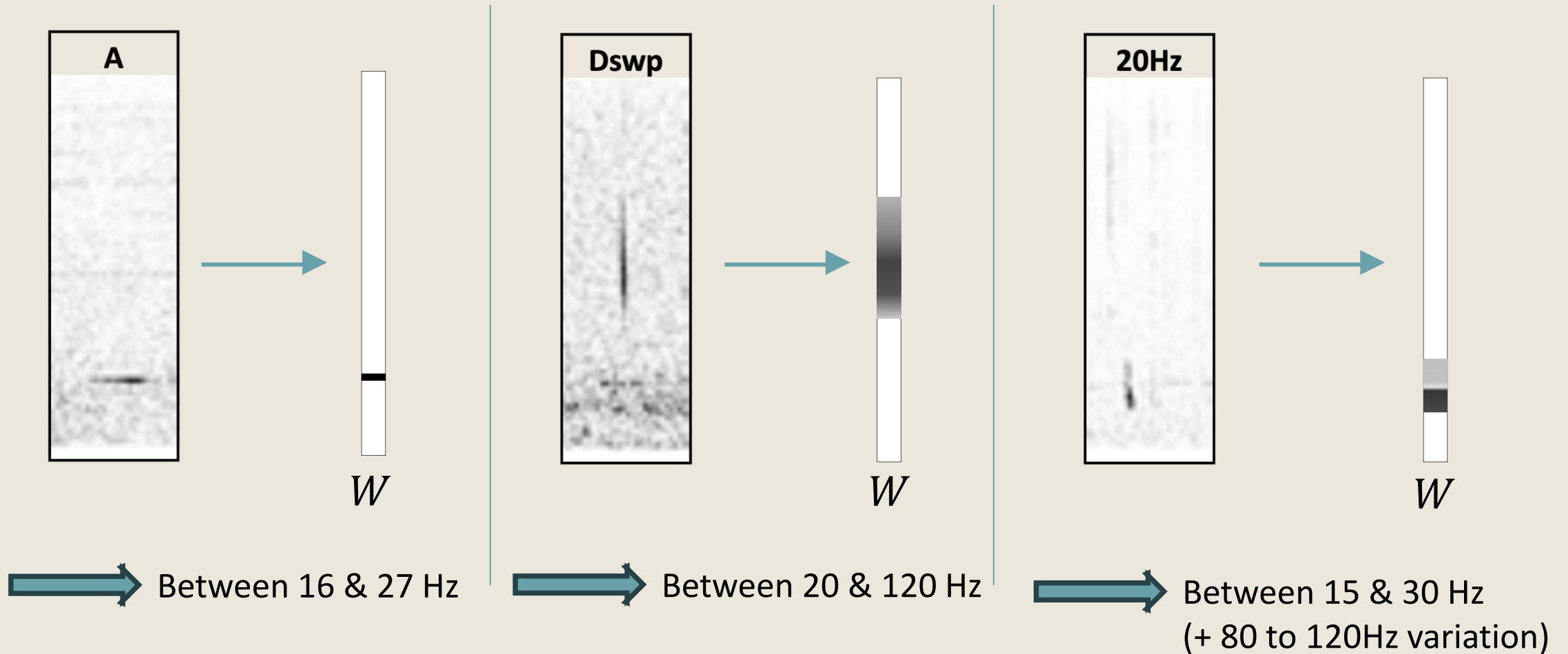
➡ Between 20 & 120 Hz



➡ Between 15 & 30 Hz
(+ 80 to 120Hz variation)

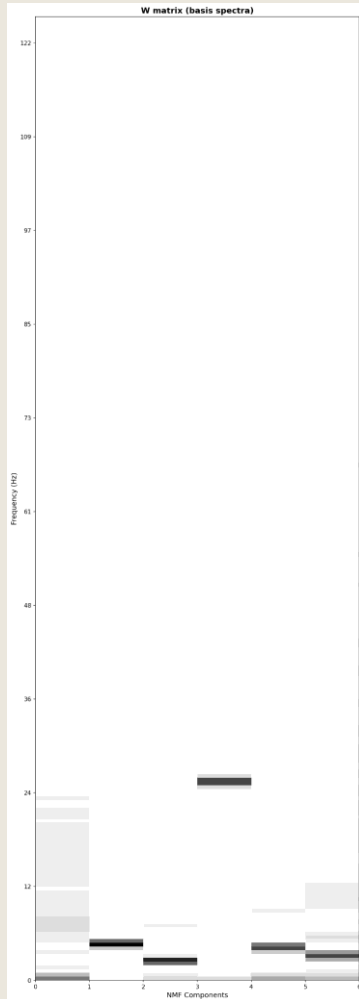
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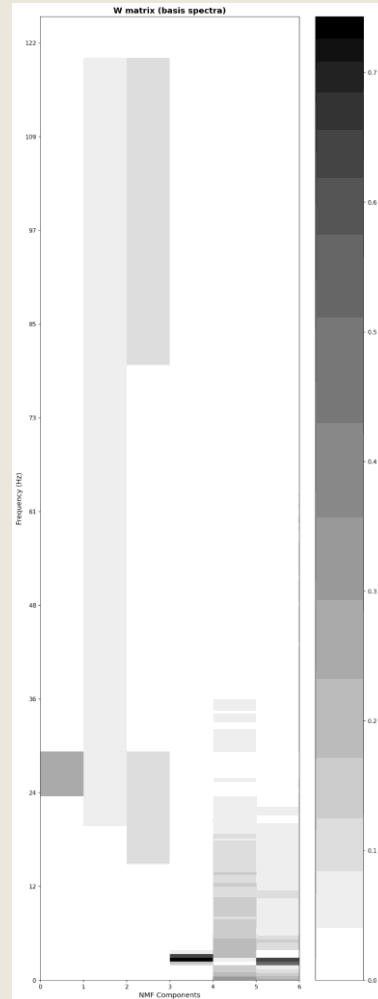


Forcing a priori information directly in W

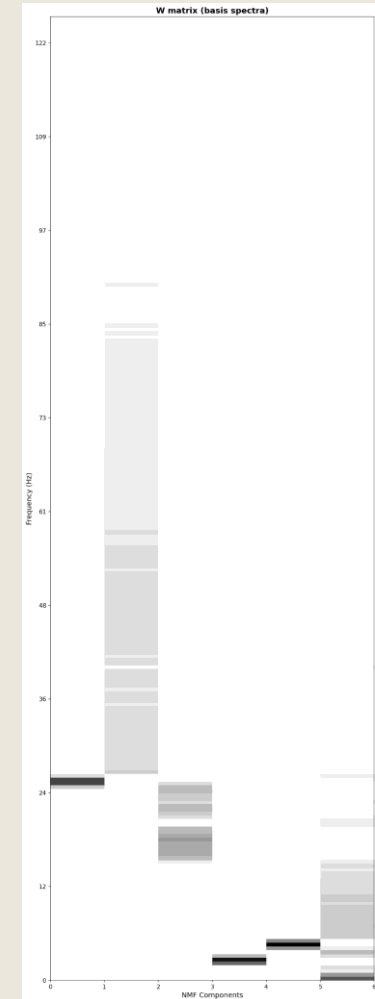
Unconstrained
(no order for the cols)



Fixed



Masked



Forcing a priori information directly in W

	ddu2021	Kerguelen2020
Unconstrained NMF (using labels)	12,50%	39,47%
NMF with fixed columns	7,29%	8,09%
NMF with masked columns	7,77%	17,77%

Comparison with BioDCASE results

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NMF masked columns					7,77%	17,77%

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Conclusions:

NMF is an **unsupervised** alternative for sound event detection in PAM



- NMF is interpretable and can be steered
- Future work could largely benefit from the existing literature
 - Contact me if you are interested!



- Levels of performance are far from SOTA
- NMF makes assumptions on the data that might be simplistic

Are NMF assumptions relevant for PAM?

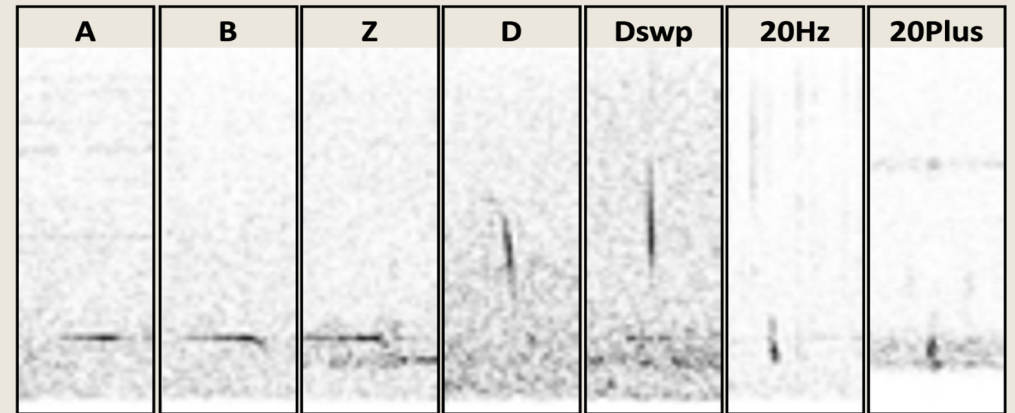
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Are NMF assumptions relevant for PAM?

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- ✓ *Ok for A, Dswp and 20Hz calls*
- ✗ *ABZ calls are simplified to A*
- ✗ *D call needs several columns*

➡ ... Meh



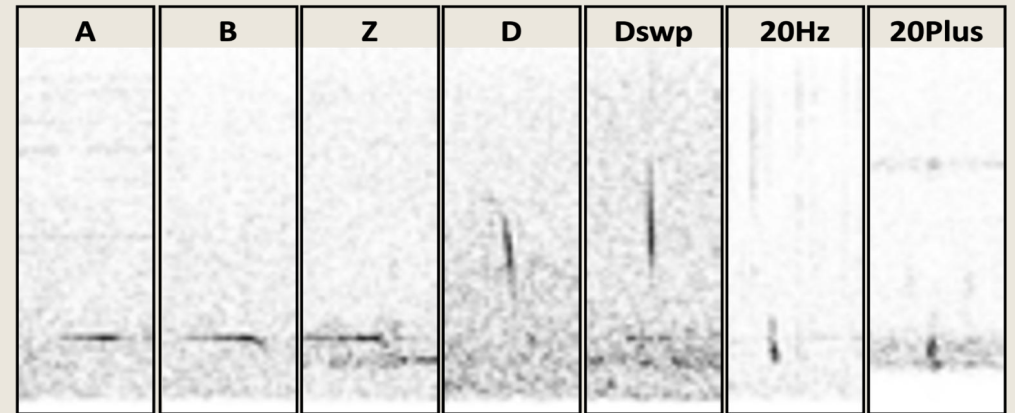
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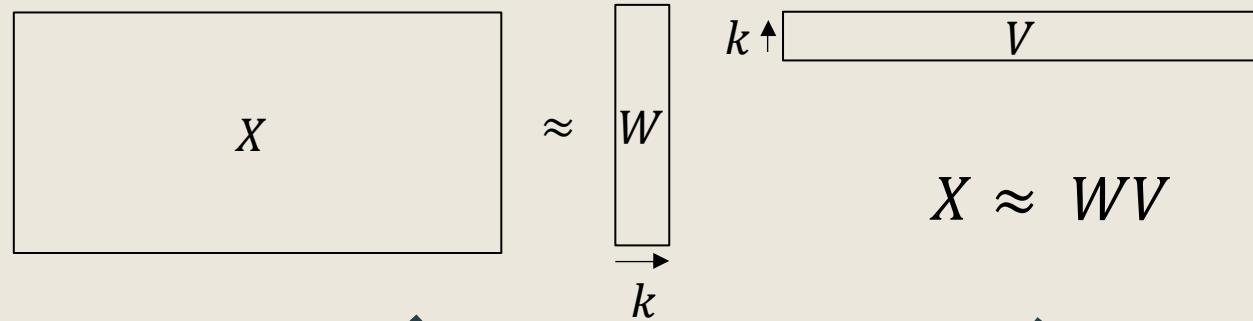
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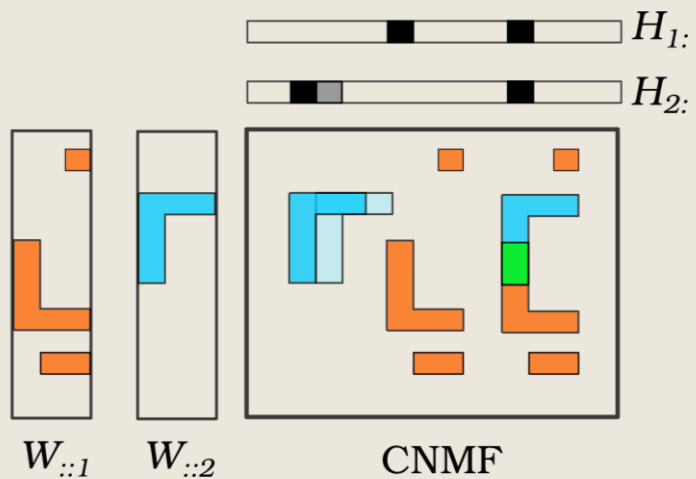
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➡ Depends a lot on the audio sample

NMF can be extended



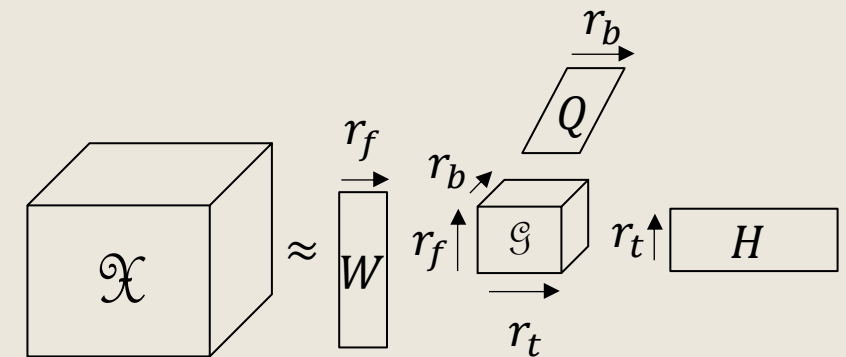
1. Convolutional NMF



2. Deep NMF

$$X \approx W_1 W_2 W_3 \dots W_L V$$

3. Tensor factorization



$$\mathcal{X} = \mathcal{G} \times_1 W \times_2 H \times_3 Q$$

UNSUPERVISED DETECTION OF WHALE CALLS BY NONNEGATIVE MATRIX FACTORIZATION



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How does NMF work?

- **Unsupervised** technique

- Mathematical problem relative to distance function $d()$:

$$\operatorname{argmin}_{W \geq 0, H \geq 0} d(X, WV)$$

- $d()$ is in general the Euclidean distance, leading to:

$$\operatorname{argmin}_{W \geq 0, H \geq 0} \|X - WV\|_2$$

- For details: [The Why and How of Nonnegative Matrix Factorization \[Gil14\]](#)

Confusion matrices

- Unconstrained NMF

	ddu2021	Kerguelen2020
bmabz	13,52% (5309 calls)	44,59% (1668 calls)
d	9,75% (4332 calls)	9,71% (200 calls)
bp	27,06% (366 calls)	18,23% (269 calls)

Confusion matrices (2)

■ Masked NMF

	ddu2021	Kerguelen2020
bmabz	11,71% (5309 calls)	26,07% (1668 calls)
d	2,34% (4332 calls)	0,83% (200 calls)
bp	4,07% (366 calls)	1,07% (269 calls)